

VERHNYAYA TOYMA, Russian Federation

WMO: 227780

Lat: 62.2412N

Lon: 45.0131E

Elev: 190

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-21.4	-33.0	0.9	-27.9	-26.4	1.3	-21.3	19.6	22.1	17.8	19.4	2.9	30	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	81.7	65.5	78.2	63.7	74.9	61.9	67.9	77.6	65.9	74.9	63.9	71.9	7.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	91.0	72.4	62.4	84.9	69.9	60.6	79.4	68.1	32.3	77.3	30.8	75.0	29.3	72.0	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.5	13.8	DB	-32.8	85.8	5.6	3.2	-36.8	88.1	-40.1	90.0	-43.2	91.8	-47.2	94.1	(4)
(5)			WB	-32.9	70.8	5.2	2.7	-36.7	72.7	-39.8	74.3	-42.7	75.9	-46.5	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	8.2	12.9	21.5	34.7	47.0	57.7	62.2	57.5	48.1	35.5	23.0	14.7		
	DBStd	21.19	15.41	15.67	10.76	9.17	9.84	7.79	6.49	6.75	6.48	7.33	11.40	14.74		
	HDD50	6368	1297	1039	883	466	184	18	2	12	111	450	811	1095		
	HDD65	10905	1762	1459	1348	909	561	243	133	247	508	914	1261	1560		
	CDD50	1026	0	0	0	7	92	249	379	245	53	1	0	0		
	CDD65	88	0	0	0	0	4	24	45	15	0	0	0	0		
	CDH74	856	0	0	0	1	52	276	401	125	1	0	0	0		
	CDH80	161	0	0	0	0	5	65	74	17	0	0	0	0		
	WSAvg	6.8	6.7	7.3	7.1	6.7	7.4	6.7	6.3	6.1	6.1	7.1	6.9	6.6		
	PrecAvg	23.1	1.5	1.3	1.0	1.4	1.8	2.2	2.4	2.6	2.2	2.4	2.0	1.9		
(15) Precipitation	PrecMax	26.2	2.1	2.3	1.7	3.0	3.0	4.6	4.9	4.7	4.4	5.5	5.0	2.8		
	PrecMin	19.8	1.0	0.4	0.2	0.4	0.7	0.6	0.8	1.1	0.9	0.4	0.8	0.9		
	PrecStd	1.8	0.3	0.5	0.4	0.8	0.6	1.1	1.3	1.1	0.9	1.2	1.0	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.7	43.9	67.8	80.0	87.0	86.8	82.6	70.6	53.1	42.7	37.3		
		MCWB	32.9	34.7	36.2	53.1	61.6	68.7	67.3	67.2	61.0	48.0	41.8	36.0		
	2%	DB	32.6	34.9	39.4	58.1	74.1	80.8	82.3	77.3	66.1	49.6	38.6	34.7		
		MCWB	31.8	33.4	34.4	47.0	57.5	64.3	66.1	64.5	57.9	47.1	37.7	33.7		
	5%	DB	30.5	33.2	36.9	51.2	69.6	76.6	78.9	73.4	62.4	47.9	36.9	33.5		
		MCWB	29.9	32.1	33.1	42.5	55.8	62.2	64.8	62.8	55.9	45.7	36.2	32.6		
	10%	DB	27.4	31.2	35.1	47.1	64.8	72.5	75.5	69.8	58.8	45.6	34.9	32.3		
		MCWB	26.7	30.2	32.3	40.0	53.2	60.0	62.6	60.9	53.8	43.5	33.9	31.5		
	0.4%	WB	33.3	35.6	37.4	52.8	64.7	70.9	71.0	68.7	62.5	49.4	42.2	36.0		
		MCDB	33.8	36.6	41.7	66.4	74.6	82.7	81.6	77.4	67.2	51.3	42.4	36.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	33.5	35.5	47.5	60.7	67.3	68.2	66.3	59.5	47.8	38.0	34.0		
		MCDB	32.7	34.5	38.4	54.7	69.2	77.2	78.4	75.3	64.7	49.5	38.5	34.4		
	5%	WB	29.9	32.2	34.0	43.5	57.4	64.6	66.3	64.2	56.7	45.9	36.4	32.9		
		MCDB	30.4	33.2	36.1	49.7	66.0	73.0	75.3	71.0	61.0	47.6	37.0	33.4		
	10%	WB	26.8	30.3	32.8	40.9	54.3	61.8	64.4	62.1	54.4	43.7	34.1	31.6		
		MCDB	27.4	31.2	34.8	46.1	64.3	70.1	73.2	68.1	58.0	45.4	34.8	32.4		
	5% DB	MDBR	11.4	11.9	16.1	15.8	18.3	18.3	17.9	16.4	14.0	8.4	7.7	9.7		
		MCDBR	8.9	8.2	14.3	22.2	25.4	25.3	23.4	22.0	20.7	10.8	6.1	7.4		
		MCWBR	8.8	7.7	10.2	13.2	12.8	11.9	9.9	10.9	13.0	9.2	5.8	7.4		
		MCWBR	8.8	8.6	10.6	19.1	21.2	21.4	20.2	19.4	17.5	10.6	6.1	7.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.8	8.3	8.0	12.2	11.7	11.3	10.2	10.5	12.1	9.6	5.8	7.6		
	taub		0.246	0.295	0.326	0.360	0.354	0.354	0.391	0.371	0.337	0.316	0.262	0.221		
		taud	1.989	2.058	2.081	2.168	2.439	2.474	2.377	2.408	2.490	2.369	2.083	1.943		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		126	200	241	259	275	277	261	256	245	201	121	75		
		Edh at Noon	12	23	33	37	31	31	33	30	23	17	11	7		
	RadAvg		75	274	689	1151	1541	1646	1653	1145	636	237	69	34		
(44) All-Sky Solar Radiation	RadStd		11	66	121	139	95	181	134	117	71	34	15	5		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page